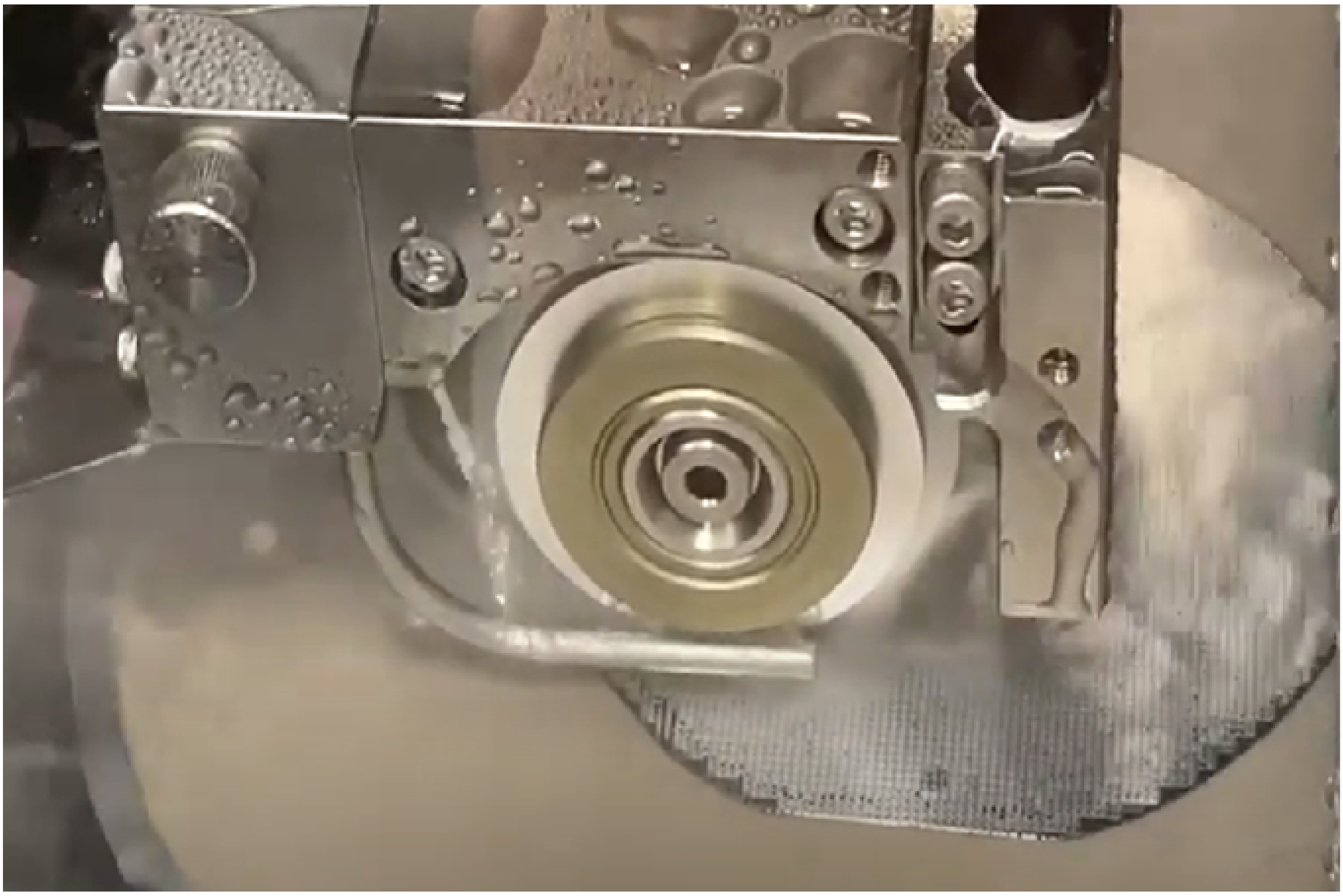
Qualification Pack



Assembly Process Supervisor- Wafer Dicing

QP Code: TEL/Q7204

Version: 1.0

NSQF Level: 5.5

Telecom || 3rd Floor, Plot No 126, Sector - 44
Gurgaon - 122003 || email:QAteam@tsscindia.com

Contents

TEL/Q7204: Assembly Process Supervisor- Wafer Dicing

Brief Job Description

The individual for this job role manages the dicing of semiconductor wafers into individual chips. The individual is also optimize dicing processes, selecting appropriate cutting tools, and ensuring the minimization of chip damage. In addition, the individual also analyzes yield data, collaborates with cross-functional teams to improve dicing strategies, and contributes to equipment maintenance protocols.

Personal Attributes

The candidate must demonstrate attention to detail, analytical and problem-solving abilities, and effective communication skills, both written and verbal. The individual should be adept at working collaboratively with others.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. TEL/N7212: Optimize Dicing Process
2. TEL/N7213: Selecting & Managing Cutting Tools
3. TEL/N7214: Yield Analysis and Improvement Strategies
4. TEL/N7215: Maintain Equipment, Records & Reports
5. DGT/VSQ/N0103: Employability Skills (90 Hours)

Qualification Pack (QP) Parameters

Sector	Telecom
Sub-Sector	Semiconductor-Manufacturing & Packaging
Occupation	Semiconductor-M&P
Country	India
NSQF Level	5.5
Credits	19
Aligned to NCO/ISCO/ISIC Code	NCO-2015/3522.9900

Minimum Educational Qualification & Experience	Graduate (3rd year of 3-years/4years UG) with NA of experience OR Completed 2nd year diploma after 12th with 2 Years of experience relevant experience OR Previous relevant Qualification of NSQF Level (5) with 1.5 years of experience relevant experience OR Previous relevant Qualification of NSQF Level (4.5) with 3 Years of experience relevant experience
Minimum Level of Education for Training in School	12th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	28/02/2026
NSQC Approval Date	22/08/2024
Version	1.0
Reference code on NQR	QG-5.5-TL-02956-2024-V1-TSSC
NQR Version	1

TEL/N7212: Optimize Dicing Process

Description

This OS unit focuses on setting up equipment, selecting blades, and iteratively refining parameters to achieve high-throughput dicing with minimal chip damage.

Scope

The scope covers the following :

- Prepare Dicing Equipment and Set Up Process
- Execute and Optimize Dicing Process

Elements and Performance Criteria

Prepare Dicing Equipment and Set Up Process

To be competent, the user/individual on the job must be able to:

PC1. review wafer specification documents (material datasheet, chip design layout) to gather relevant information

PC2. identify relevant wafer parameters (material composition, thickness, desired chip dimensions) for dicing

PC3. select appropriate dicing blade type (e.g., diamond, abrasive) based on wafer material and desired chip edge quality (smoothness, minimal chipping)

PC4. consult reference tables or software to determine initial dicing parameter settings (speed, force, etc.) considering wafer properties and blade selection

PC5. prepare the dicing equipment work area according to SOPs (cleanliness, organization)

PC6. mount the wafer securely onto the dicing stage using appropriate holding fixtures as per SOPs

PC7. install the selected dicing blade following safe handling procedures to avoid injury

PC8. perform calibration procedures for dicing equipment components (stage movement accuracy, blade tension, vibration control) as per manufacturer's instructions

PC9. verify calibration accuracy using reference standards (e.g., gauge blocks)

Execute and Optimize Dicing Process

To be competent, the user/individual on the job must be able to:

PC10. run initial test dicing cycles with pre-determined parameters

PC11. inspect diced wafers for chip damage (cracking, chipping) and edge quality

PC12. analyze process data (throughput, cycle time) collected during test runs

PC13. • adjust dicing parameters (speed, force, blade selection if necessary), based on analysis, to achieve:
• a. high throughput (maximize wafer processing rate)
• b. minimal chip damage (maintain chip quality)

PC14. repeat test dicing cycles with adjusted parameters and continue iteratively optimizing for best results

PC15. monitor dicing process parameters and equipment performance during operation continuously

PC16. record dicing process data (parameters used, yield results, cycle time, blade wear indicators) in designated formats (e.g., logs, electronic records)

PC17. analyze recorded data to identify trends, potential areas for further optimization, and opportunities for preventive maintenance (based on blade wear data)

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. different types of serials (e.g., silicon, silicon carbide) and their properties (hardness, brittleemiconductor wafer matness)

KU2. impact of wafer thickness on dicing parameters and chip quality

KU3. interpretation of wafer design layout for chip dimensions and spacing requirements

KU4. principles of dicing blade selection based on wafer material and desired chip edge quality (smoothness, minimal chipping)

KU5. different types of dicing blades (e.g., diamond, abrasive) and their characteristics

KU6. factors influencing blade selection, such as cost-effectiveness and compatibility with dicing equipment

KU7. safe handling procedures for dicing blades

KU8. proper use of dicing equipment controls for stage movement, blade tension, and vibration control

KU9. standard operating procedures (SOPs) for dicing equipment setup and calibration

KU10. functionality and purpose of dicing equipment components (stage, blade holder, vibration dampener)

KU11. calibration procedures for dicing equipment components using reference standards (e.g., gauge blocks)

KU12. importance of accurate record keeping during dicing equipment setup and calibration.

KU13. definition and influence of key dicing parameters (speed, force, blade selection) on throughput, chip quality, and blade wear

KU14. techniques for visual inspection of diced wafers to identify chip damage (cracking, chipping) and edge quality issues

KU15. interpretation of process data (throughput, cycle time) to assess dicing efficiency

KU16. use of data analysis to identify areas for improvement in the dicing process

KU17. principles of iterative optimization for balancing high throughput with minimal chip damage

KU18. techniques for adjusting dicing parameters based on analysis of process data and wafer inspection results

KU19. importance of continuous monitoring of dicing process parameters and equipment performance

KU20. selection of appropriate data points to record during the dicing process (parameters used, yield results, cycle time, blade wear indicators)

KU21. use of designated formats (logs, electronic records) for accurate and consistent data recording

KU22. correlation between blade wear data and potential equipment maintenance needs

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. maintain work-related notes and records
GS2. read the relevant literature to get the latest updates about the field of work
GS3. listen attentively to understand the information/ instructions being shared
GS4. communicate politely and professionally
GS5. plan and prioritize tasks to ensure timely completion
GS6. coordinate with the co-workers to achieve the work objectives
GS7. evaluate all possible solutions to a problem to select the best one
GS8. take quick decisions to deal with workplace emergencies/ accidents

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Prepare Dicing Equipment and Set Up Process	**14**	**30**	**-**	**5**
PC1. review wafer specification documents (material datasheet, chip design layout) to gather relevant information	2	3	-	1
PC2. identify relevant wafer parameters (material composition, thickness, desired chip dimensions) for dicing	1	3	-	-
PC3. select appropriate dicing blade type (e.g., diamond, abrasive) based on wafer material and desired chip edge quality (smoothness, minimal chipping)	1	3	-	1
PC4. consult reference tables or software to determine initial dicing parameter settings (speed, force, etc.) considering wafer properties and blade selection	1	3	-	1
PC5. prepare the dicing equipment work area according to SOPs (cleanliness, organization)	2	3	-	-
PC6. mount the wafer securely onto the dicing stage using appropriate holding fixtures as per SOPs	1	3	-	-
PC7. install the selected dicing blade following safe handling procedures to avoid injury	3	5	-	1
PC8. perform calibration procedures for dicing equipment components (stage movement accuracy, blade tension, vibration control) as per manufacturer's instructions	2	4	-	-
PC9. verify calibration accuracy using reference standards (e.g., gauge blocks)	1	3	-	1
Execute and Optimize Dicing Process	**16**	**30**	**-**	**5**
PC10. run initial test dicing cycles with pre-determined parameters	3	5	-	1
PC11. inspect diced wafers for chip damage (cracking, chipping) and edge quality	2	3	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. analyze process data (throughput, cycle time) collected during test runs	2	4	-	-
PC13. • adjust dicing parameters (speed, force, blade selection if necessary), based on analysis, to achieve: • a. high throughput (maximize wafer processing rate) • b. minimal chip damage (maintain chip quality)	3	6	-	2
PC14. repeat test dicing cycles with adjusted parameters and continue iteratively optimizing for best results	1	3	-	-
PC15. monitor dicing process parameters and equipment performance during operation continuously	1	3	-	-
PC16. record dicing process data (parameters used, yield results, cycle time, blade wear indicators) in designated formats (e.g., logs, electronic records)	2	3	-	1
PC17. analyze recorded data to identify trends, potential areas for further optimization, and opportunities for preventive maintenance (based on blade wear data)	2	3	-	1
NOS Total	**30**	**60**	**-**	**10**

National Occupational Standards (NOS) Parameters

NOS Code	TEL/N7212
NOS Name	Optimize Dicing Process
Sector	Telecom
Sub-Sector	
Occupation	Semiconductor-M&P
NSQF Level	5.5
Credits	6
Version	1.0
Last Reviewed Date	22/08/2024
Next Review Date	22/08/2025
NSQC Clearance Date	22/08/2024

TEL/N7213: Selecting & Managing Cutting Tools

Description

This OS unit ensures optimal dicing by selecting suitable blades and maintaining them through inspection and inventory control.

Scope

The scope covers the following :

- Select Appropriate Dicing Blades
- Dicing Blade Maintenance and Inventory Management

Elements and Performance Criteria

Select Appropriate Dicing Blades

To be competent, the user/individual on the job must be able to:

PC1. review wafer specification documents (material datasheet, chip design layout) to obtain key information

PC2. determine the wafer material composition (e.g., silicon, sapphire) and thickness from the specifications

PC3. identify the desired chip size (dimensions) from the chip design layout

PC4. determine the required edge quality for the diced chips (e.g., smoothness, minimal chipping) based on application needs

PC5. consult blade selection guide or manufacturer recommendations based on identified wafer material and desired chip characteristics

PC6. select appropriate dicing blade type (e.g., diamond, abrasive) considering compatibility, cost-effectiveness, and desired cutting performance

Dicing Blade Maintenance and Inventory Management

To be competent, the user/individual on the job must be able to:

PC7. establish a routine inspection schedule for dicing blades as per manufacturer's recommendations or company SOPs

PC8. visually inspect blades for any signs of wear and tear that may affect cutting performance

PC9. utilize appropriate inspection tools (e.g., magnifying glass, blade wear gauge) for detailed examination

PC10. document inspection findings and blade condition (usable, requires replacement)

PC11. establish minimum and maximum inventory levels for different types of dicing blades based on usage patterns and lead times for procurement

PC12. monitor current blade inventory levels (quantity, type) through a designated tracking system (e.g., inventory management software, physical inventory checks)

PC13. initiate blade procurement processes (purchase orders) when inventory levels fall below the minimum threshold to ensure sufficient stock before depletion

PC14. maintain proper storage conditions for dicing blades according to manufacturer's recommendations (e.g., humidity control, dust-free environment)

PC15. properly dispose of used or worn-out blades following established safety and environmental regulations

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. different types of semiconductor wafer materials (e.g., silicon, silicon carbide) and their properties relevant to dicing (hardness, brittleness)
KU2. impact of wafer thickness on blade selection and potential chip edge quality issues
KU3. interpretation of wafer design layout for chip dimensions and desired edge quality (smoothness, minimal chipping)
KU4. principles of dicing blade selection based on wafer material and desired chip edge quality
KU5. different types of dicing blades (e.g., diamond, abrasive) and their cutting characteristics (durability, smoothness of cut)
KU6. factors influencing blade selection, such as compatibility with dicing equipment and cost-effectiveness
KU7. ability to utilize blade selection guides and manufacturer recommendations for blade selection based on specific wafer material and desired chip characteristics
KU8. signs of wear and tear on dicing blades (chipped segments, reduced diameter, blade glazing)
KU9. proper use of inspection tools (magnifying glass, blade wear gauge) for detailed blade examination
KU10. importance of documenting inspection findings and blade condition (usable, requires replacement) for tracking purposes
KU11. principles of inventory management, including establishing minimum and maximum inventory levels based on usage patterns and lead times
KU12. techniques for monitoring current blade inventory levels (quantity, type) through designated tracking systems
KU13. procedures for initiating blade procurement processes (purchase orders) to maintain sufficient stock and avoid production stoppages
KU14. proper storage conditions for dicing blades (humidity control, dust-free environment) to maintain optimal performance and lifespan
KU15. safe and environmentally responsible disposal procedures for used or worn-out dicing blades

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. maintain work related records
GS2. read the relevant guides and literature to get the latest information about the field of work
GS3. communicate clearly and politely
GS4. listen attentively to understand the information/instructions being shared
GS5. plan and prioritize tasks to ensure timely completion
GS6. identify appropriate solutions to work related issues

GS7. take quick decisions in case of an emergency/accident

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Select Appropriate Dicing Blades	**12**	**30**	**-**	**5**
PC1. review wafer specification documents (material datasheet, chip design layout) to obtain key information	2	5	-	1
PC2. determine the wafer material composition (e.g., silicon, sapphire) and thickness from the specifications	3	6	-	1
PC3. identify the desired chip size (dimensions) from the chip design layout	1	4	-	1
PC4. determine the required edge quality for the diced chips (e.g., smoothness, minimal chipping) based on application needs	1	4	-	-
PC5. consult blade selection guide or manufacturer recommendations based on identified wafer material and desired chip characteristics	3	6	-	1
PC6. select appropriate dicing blade type (e.g., diamond, abrasive) considering compatibility, cost-effectiveness, and desired cutting performance	2	5	-	1
Dicing Blade Maintenance and Inventory Management	**18**	**30**	**-**	**5**
PC7. establish a routine inspection schedule for dicing blades as per manufacturer's recommendations or company SOPs	2	3	-	1
PC8. visually inspect blades for any signs of wear and tear that may affect cutting performance	1	3	-	-
PC9. utilize appropriate inspection tools (e.g., magnifying glass, blade wear gauge) for detailed examination	2	4	-	1
PC10. document inspection findings and blade condition (usable, requires replacement)	3	3	-	-
PC11. establish minimum and maximum inventory levels for different types of dicing blades based on usage patterns and lead times for procurement	2	3	-	1

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. monitor current blade inventory levels (quantity, type) through a designated tracking system (e.g., inventory management software, physical inventory checks)	3	4	-	1
PC13. initiate blade procurement processes (purchase orders) when inventory levels fall below the minimum threshold to ensure sufficient stock before depletion	2	3	-	-
PC14. maintain proper storage conditions for dicing blades according to manufacturer's recommendations (e.g., humidity control, dust-free environment)	1	3	-	1
PC15. properly dispose of used or worn-out blades following established safety and environmental regulations	2	4	-	-
NOS Total	**30**	**60**	**-**	**10**

National Occupational Standards (NOS) Parameters

NOS Code	TEL/N7213
NOS Name	Selecting & Managing Cutting Tools
Sector	Telecom
Sub-Sector	
Occupation	Semiconductor-M&P
NSQF Level	5.5
Credits	4
Version	1.0
Last Reviewed Date	22/08/2024
Next Review Date	22/08/2025
NSQC Clearance Date	22/08/2024

TEL/N7214: Yield Analysis and Improvement Strategies

Description

This OS unit tackles dicing defects by analyzing yield data, collaborating on solutions, and implementing corrective actions to boost wafer yield.

Scope

The scope covers the following :

- Dicing Yield Analysis & Strategy Development
- Implementation & Monitoring of Corrective Actions

Elements and Performance Criteria

Dicing Yield Analysis & Strategy Development

To be competent, the user/individual on the job must be able to:

PC1. collect diced wafer yield data from various sources (e.g., process monitoring systems, inspection reports)

PC2. identify patterns and trends in defect types observed in the yield data (e.g., chipping, cracking, contamination)

PC3. analyze the frequency and distribution of each defect type across different batches or wafers

PC4. compare yield data with process parameters and equipment logs from corresponding dicing jobs

PC5. identify potential correlations between specific process settings (speed, force, blade type) and observed defect types

PC6. prioritize yield issues based on their severity and impact on overall yield.

PC7. initiate discussions with cross-functional teams (process engineers, quality control) to share yield data and defect analysis

PC8. brainstorm and propose potential solutions for identified yield issues based on expertise of each team

PC9. evaluate proposed solutions considering factors like feasibility, cost-effectiveness, and potential impact on other process parameters

PC10. develop and document a collaborative action plan for yield improvement

PC11. define clear tasks, responsibilities, and timelines for implementing the chosen strategies

Implementation & Monitoring of Corrective Actions

To be competent, the user/individual on the job must be able to:

PC12. implement the agreed-upon corrective actions for yield improvement (e.g., adjust process parameters, modify equipment settings, implement new cleaning procedures)

PC13. ensure clear communication and coordination with personnel responsible for implementation

PC14. collect new yield data after implementing the corrective actions

PC15. monitor the impact of implemented changes on defect occurrence and overall yield percentage

PC16. assess the effectiveness of the corrective actions based on the collected data
PC17. identify if further adjustments or modifications are necessary to achieve optimal yield
PC18. document the results of the implemented actions and their impact on yield improvement
PC19. share the information with relevant teams to facilitate continuous yield improvement and knowledge sharing

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. dicing process steps and how they can impact wafer yield
KU2. how to interpret process data and equipment logs to identify potential causes of yield issues
KU3. techniques for analyzing diced wafer yield data to identify trends and patterns in defect types
KU4. statistical methods for prioritizing yield issues based on their severity and impact on overall yield
KU5. recognition of common dicing defects (chipping, cracking, surface contamination) from inspection reports and wafer samples
KU6. correlation between defect types and potential causes in the dicing process
KU7. importance of clear communication and collaboration with different teams (process engineers, quality control) to share data and expertise
KU8. how to participate in brainstorming sessions to develop solutions for yield improvement
KU9. strategies for improving dicing yield, such as adjusting process parameters, modifying equipment settings, and implementing new cleaning procedures
KU10. how to interpret post-implementation yield data to assess the effectiveness of corrective actions
KU11. importance of documenting and sharing improvement results with relevant teams to facilitate ongoing yield optimization

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. maintain work related records
GS2. read the relevant guides and literature to get the latest information about the field of work
GS3. communicate clearly and politely
GS4. listen attentively to understand the information/instructions being shared
GS5. plan and prioritize tasks to ensure timely completion
GS6. identify appropriate solutions to work related issues
GS7. take quick decisions in case of an emergency/accident

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Dicing Yield Analysis & Strategy Development	**16**	**30**	**-**	**5**
PC1. collect diced wafer yield data from various sources (e.g., process monitoring systems, inspection reports)	2	4	-	1
PC2. identify patterns and trends in defect types observed in the yield data (e.g., chipping, cracking, contamination)	2	2	-	-
PC3. analyze the frequency and distribution of each defect type across different batches or wafers	1	2	-	-
PC4. compare yield data with process parameters and equipment logs from corresponding dicing jobs	2	4	-	1
PC5. identify potential correlations between specific process settings (speed, force, blade type) and observed defect types	2	2	-	-
PC6. prioritize yield issues based on their severity and impact on overall yield.	1	2	-	-
PC7. initiate discussions with cross-functional teams (process engineers, quality control) to share yield data and defect analysis	2	4	-	1
PC8. brainstorm and propose potential solutions for identified yield issues based on expertise of each team	1	3	-	1
PC9. evaluate proposed solutions considering factors like feasibility, cost-effectiveness, and potential impact on other process parameters	1	2	-	-
PC10. develop and document a collaborative action plan for yield improvement	1	3	-	1
PC11. define clear tasks, responsibilities, and timelines for implementing the chosen strategies	1	2	-	-
Implementation & Monitoring of Corrective Actions	**14**	**30**	**-**	**5**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. implement the agreed-upon corrective actions for yield improvement (e.g., adjust process parameters, modify equipment settings, implement new cleaning procedures)	2	4	-	1
PC13. ensure clear communication and coordination with personnel responsible for implementation	1	3	-	-
PC14. collect new yield data after implementing the corrective actions	2	4	-	1
PC15. monitor the impact of implemented changes on defect occurrence and overall yield percentage	2	5	-	1
PC16. assess the effectiveness of the corrective actions based on the collected data	2	4	-	-
PC17. identify if further adjustments or modifications are necessary to achieve optimal yield	2	4	-	1
PC18. document the results of the implemented actions and their impact on yield improvement	2	3	-	1
PC19. share the information with relevant teams to facilitate continuous yield improvement and knowledge sharing	1	3	-	-
NOS Total	**30**	**60**	**-**	**10**

National Occupational Standards (NOS) Parameters

NOS Code	TEL/N7214
NOS Name	Yield Analysis and Improvement Strategies
Sector	Telecom
Sub-Sector	
Occupation	Semiconductor-M&P
NSQF Level	5.5
Credits	3
Version	1.0
Last Reviewed Date	22/08/2024
Next Review Date	22/08/2025
NSQC Clearance Date	22/08/2024

TEL/N7215: Maintain Equipment, Records & Reports

Description

This OS unit ensures smooth operation through equipment maintenance, accurate data recording, insightful reporting, and strict adherence to safety protocols.

Scope

The scope covers the following :

- Perform Preventive Maintenance & Calibration
- Maintain Up-to-Date Records & Reports
- Comply with Safety Protocols

Elements and Performance Criteria

Perform Preventive Maintenance & Calibration

To be competent, the user/individual on the job must be able to:

PC1. review manufacturer's recommended maintenance schedule for dicing equipment

PC2. identify and locate specific components requiring maintenance (e.g., bearings, filters, blades)

PC3. perform basic cleaning tasks on the dicing equipment as per guidelines (e.g., dust removal, debris cleaning)

PC4. replenish or replace consumables according to the schedule (e.g., lubricants, coolants).

PC5. utilize knowledge of typical dicing equipment operation sounds and vibration levels to differentiate between normal operation and potential issues

PC6. report any unusual observations during maintenance activities, such as excessive wear, loose components, or strange noises

PC7. assist qualified personnel during scheduled calibration procedures for dicing equipment components

PC8. identify the purpose and function of each calibration step being performed (e.g., stage movement calibration ensures precise blade positioning)

PC9. record calibration data (measurements, adjustments made) on designated forms or electronic systems

PC10. report any discrepancies or unexpected results observed during calibration to the qualified personnel

PC11. maintain records of calibration activities, including the date, equipment components calibrated, and any relevant observations

Maintain Up-to-Date Records & Reports

To be competent, the user/individual on the job must be able to:

PC12. record dicing process parameters (speed, force, blade type) in designated formats (e.g., logs, electronic data entry) during dicing operation

PC13. collect and document yield data (good/defect counts) after dicing process completion

PC14. maintain accurate records of equipment maintenance activities (date, tasks) performed

PC15. regularly review recorded data for accuracy and completeness

PC16. update records promptly to reflect any changes or corrections identified

PC17. prepare reports summarizing dicing process performance metrics (throughput, yield percentage)

PC18. analyze yield data to identify trends and potential areas for improvement

PC19. generate reports on equipment maintenance and any corrective actions

PC20. submit reports on dicing process performance following established channels

PC21. review and thoroughly understand company policies and procedures related to documentation and record-keeping for dicing operations

PC22. follow company-defined formats and procedures for recording dicing process data and generating reports

PC23. ensure data entries are clear, concise, and verifiable by others

PC24. properly store and archive dicing process records according to company guidelines

Comply with Safety Protocols

To be competent, the user/individual on the job must be able to:

PC25. follow established safety protocols for operating dicing equipment (e.g., lockout/tagout procedures, proper blade handling)

PC26. identify and adhere to safety regulations for handling hazardous materials potentially used during dicing (e.g., coolants, cleaning solutions)

PC27. identify potential safety hazards (e.g., electrical risks, slipping hazards) and be observant of the dicing work environment

PC28. report identified safety hazards to supervisors or designated personnel for corrective action

PC29. wear appropriate PPE (e.g., safety glasses, gloves, ear protection) as mandated by company safety regulations for dicing operations

PC30. ensure PPE is properly fitted, maintained, and in good working condition

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. manufacturer's recommended maintenance schedules for dicing equipment

KU2. basic cleaning and lubrication procedures for dicing equipment components

KU3. importance of calibration for maintaining consistent dicing performance

KU4. how to identify unusual observations during maintenance (e.g., wear, noise)

KU5. assisting qualified personnel with calibration procedures

KU6. recording calibration data on designated forms

KU7. dicing process parameters affecting results (speed, force, blade type)

KU8. methods for collecting and documenting yield data (good/defect counts)

KU9. analyzing yield data to identify trends and potential issues

KU10. company procedures for recording data, generating reports, and storing records

KU11. safe operating procedures for dicing equipment (lockout/tagout, blade handling)

KU12. regulations for handling hazardous materials used during dicing (coolants, cleaning solutions)

KU13. how to identify potential safety hazards in the dicing workplace (electrical, slipping)
KU14. proper use and maintenance of personal protective equipment (PPE)

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. maintain work related records
GS2. read the relevant guides and literature to get the latest information about the field of work
GS3. communicate clearly and politely
GS4. listen attentively to understand the information/instructions being shared
GS5. plan and prioritize tasks to ensure timely completion
GS6. identify appropriate solutions to work related issues
GS7. take quick decisions in case of an emergency/accident

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Perform Preventive Maintenance & Calibration	**18**	**24**	**-**	**4**
PC1. review manufacturer's recommended maintenance schedule for dicing equipment	1	1	-	1
PC2. identify and locate specific components requiring maintenance (e.g., bearings, filters, blades)	2	2	-	-
PC3. perform basic cleaning tasks on the dicing equipment as per guidelines (e.g., dust removal, debris cleaning)	2	3	-	-
PC4. replenish or replace consumables according to the schedule (e.g., lubricants, coolants).	1	2	-	-
PC5. utilize knowledge of typical dicing equipment operation sounds and vibration levels to differentiate between normal operation and potential issues	2	3	-	1
PC6. report any unusual observations during maintenance activities, such as excessive wear, loose components, or strange noises	2	2	-	-
PC7. assist qualified personnel during scheduled calibration procedures for dicing equipment components	1	2	-	-
PC8. identify the purpose and function of each calibration step being performed (e.g., stage movement calibration ensures precise blade positioning)	2	2	-	1
PC9. record calibration data (measurements, adjustments made) on designated forms or electronic systems	2	3	-	-
PC10. report any discrepancies or unexpected results observed during calibration to the qualified personnel	2	2	-	1
PC11. maintain records of calibration activities, including the date, equipment components calibrated, and any relevant observations	1	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Maintain Up-to-Date Records & Reports	**14**	**14**	**-**	**3**
PC12. record dicing process parameters (speed, force, blade type) in designated formats (e.g., logs, electronic data entry) during dicing operation	2	2	-	1
PC13. collect and document yield data (good/defect counts) after dicing process completion	1	1	-	-
PC14. maintain accurate records of equipment maintenance activities (date, tasks) performed	1	1	-	1
PC15. regularly review recorded data for accuracy and completeness	1	1	-	-
PC16. update records promptly to reflect any changes or corrections identified	1	1	-	-
PC17. prepare reports summarizing dicing process performance metrics (throughput, yield percentage)	1	1	-	-
PC18. analyze yield data to identify trends and potential areas for improvement	1	1	-	-
PC19. generate reports on equipment maintenance and any corrective actions	1	1	-	1
PC20. submit reports on dicing process performance following established channels	1	1	-	-
PC21. review and thoroughly understand company policies and procedures related to documentation and record-keeping for dicing operations	1	1	-	-
PC22. follow company-defined formats and procedures for recording dicing process data and generating reports	1	1	-	-
PC23. ensure data entries are clear, concise, and verifiable by others	1	1	-	-
PC24. properly store and archive dicing process records according to company guidelines	1	1	-	-
Comply with Safety Protocols	**8**	**12**	**-**	**3**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC25. follow established safety protocols for operating dicing equipment (e.g., lockout/tagout procedures, proper blade handling)	2	3	-	1
PC26. identify and adhere to safety regulations for handling hazardous materials potentially used during dicing (e.g., coolants, cleaning solutions)	2	2	-	-
PC27. identify potential safety hazards (e.g., electrical risks, slipping hazards) and be observant of the dicing work environment	1	2	-	1
PC28. report identified safety hazards to supervisors or designated personnel for corrective action	1	2	-	-
PC29. wear appropriate PPE (e.g., safety glasses, gloves, ear protection) as mandated by company safety regulations for dicing operations	1	2	-	1
PC30. ensure PPE is properly fitted, maintained, and in good working condition	1	1	-	-
NOS Total	**40**	**50**	**-**	**10**

National Occupational Standards (NOS) Parameters

NOS Code	TEL/N7215
NOS Name	Maintain Equipment, Records & Reports
Sector	Telecom
Sub-Sector	
Occupation	Semiconductor-M&P
NSQF Level	5.5
Credits	3
Version	1.0
Last Reviewed Date	22/08/2024
Next Review Date	22/08/2025
NSQC Clearance Date	22/08/2024

DGT/VSQ/N0103: Employability Skills (90 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the current job market requirement and future of work

PC2. identify and explore learning and employability relevant portals

PC3. research about the different industries, job market trends, latest skills required and the available opportunities

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC4. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.

PC5. follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC6. recognize the significance of 21st Century Skills for employment

PC7. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

PC8. adopt a continuous learning mindset for personal and professional development

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC9. use basic English for everyday conversation in different contexts, in person and over the telephone

PC10. read and understand routine information, notes, instructions, mails, letters etc. written in English

PC11. write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

PC12. identify career goals based on the skills, interests, knowledge, and personal attributes

PC13. prepare a career development plan with short- and long-term goals

Communication Skills

To be competent, the user/individual on the job must be able to:

PC14. follow verbal and non-verbal communication etiquette while communicating in professional and public settings

PC15. use active listening techniques for effective communication

PC16. communicate in writing using appropriate style and format based on formal or informal requirements

PC17. work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC18. communicate and behave appropriately with all genders and PwD

PC19. escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC20. identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.

PC21. carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook

PC22. identify common components of salary and compute income, expenses, taxes, investments etc

PC23. identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC24. operate digital devices and use their features and applications securely and safely

PC25. carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.

PC26. display responsible online behaviour while using various social media platforms

PC27. create a personal email account, send and process received messages as per requirement

PC28. carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications

PC29. utilize virtual collaboration tools to work effectively

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC30. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research

PC31. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion

PC32. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

PC33. identify different types of customers and ways to communicate with them

PC34. identify and respond to customer requests and needs in a professional manner

PC35. use appropriate tools to collect customer feedback

PC36. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC37. create a professional Curriculum vitae (Résumé)

PC38. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC39. apply to identified job openings using offline /online methods as per requirement

PC40. answer questions politely, with clarity and confidence, during recruitment and selection

PC41. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. components of salary and how to compute income and expenditure
KU12. importance of maintaining safety and security in offline and online financial transactions
KU13. different legal rights and laws
KU14. different types of digital devices and the procedure to operate them safely and securely
KU15. how to create and operate an e- mail account
KU16. use applications such as word processors, spreadsheets etc.
KU17. how to identify business opportunities
KU18. types and needs of customers
KU19. how to apply for a job and prepare for an interview
KU20. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence in English and other languages
GS2. communicate effectively using appropriate language in formal and informal settings
GS3. behave politely and appropriately with all to maintain effective work relationship
GS4. how to work in a virtual mode, using various technological platforms
GS5. perform calculations efficiently
GS6. solve problems effectively
GS7. pay attention to details
GS8. manage time efficiently
GS9. maintain hygiene and sanitization to avoid infection

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Introduction to Employability Skills	**1**	**1**	**-**	**-**
PC1. understand the significance of employability skills in meeting the current job market requirement and future of work	-	-	-	-
PC2. identify and explore learning and employability relevant portals	-	-	-	-
PC3. research about the different industries, job market trends, latest skills required and the available opportunities	-	-	-	-
Constitutional values – Citizenship	**1**	**1**	**-**	**-**
PC4. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC5. follow environmentally sustainable practices	-	-	-	-
Becoming a Professional in the 21st Century	**1**	**3**	**-**	**-**
PC6. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC7. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
PC8. adopt a continuous learning mindset for personal and professional development	-	-	-	-
Basic English Skills	**3**	**4**	**-**	**-**
PC9. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC11. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
Career Development & Goal Setting	**1**	**2**	**-**	**-**
PC12. identify career goals based on the skills, interests, knowledge, and personal attributes	-	-	-	-
PC13. prepare a career development plan with short- and long-term goals	-	-	-	-
Communication Skills	**2**	**2**	**-**	**-**
PC14. follow verbal and non-verbal communication etiquette while communicating in professional and public settings	-	-	-	-
PC15. use active listening techniques for effective communication	-	-	-	-
PC16. communicate in writing using appropriate style and format based on formal or informal requirements	-	-	-	-
PC17. work collaboratively with others in a team	-	-	-	-
Diversity & Inclusion	**1**	**1**	**-**	**-**
PC18. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC19. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
Financial and Legal Literacy	**2**	**3**	**-**	**-**
PC20. identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.	-	-	-	-
PC21. carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC23. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
Essential Digital Skills	**3**	**5**	**-**	**-**
PC24. operate digital devices and use their features and applications securely and safely	-	-	-	-
PC25. carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.	-	-	-	-
PC26. display responsible online behaviour while using various social media platforms	-	-	-	-
PC27. create a personal email account, send and process received messages as per requirement	-	-	-	-
PC28. carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications	-	-	-	-
PC29. utilize virtual collaboration tools to work effectively	-	-	-	-
Entrepreneurship	**2**	**3**	**-**	**-**
PC30. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC31. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC32. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
Customer Service	**1**	**2**	**-**	**-**
PC33. identify different types of customers and ways to communicate with them	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC34. identify and respond to customer requests and needs in a professional manner	-	-	-	-
PC35. use appropriate tools to collect customer feedback	-	-	-	-
PC36. follow appropriate hygiene and grooming standards	-	-	-	-
Getting ready for apprenticeship & Jobs	**2**	**3**	**-**	**-**
PC37. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC38. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC39. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC40. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC41. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	**20**	**30**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0103
NOS Name	Employability Skills (90 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/PC.

2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.

3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.

4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).

5. Individual assessment agencies will create unique evaluations for skill practical for every student at

each examination/ training center based on these criteria.

6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass 70 % aggregate for the QP.

7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(**Please note**: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
TEL/N7212.Optimize Dicing Process	30	60	-	10	100	20
TEL/N7213.Selecting & Managing Cutting Tools	30	60	-	10	100	25
TEL/N7214.Yield Analysis and Improvement Strategies	30	60	-	10	100	25
TEL/N7215.Maintain Equipment, Records & Reports	40	50	-	10	100	20
DGT/VSQ/N0103.Employability Skills (90 Hours)	20	30	-	-	50	10
Total	**150**	**260**	**-**	**40**	**450**	**100**

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.